

HARP: Human-Aware Routing Policies for Multi-Stage Decision Pipelines

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Abstract—A concise summary of the motivation, approach, and main findings of the work. Avoid citations and equations. Typically 150–250 words.

Keywords— keyword1, keyword2, keyword3, keyword4

I. INTRODUCTION

High-stakes institutional decision like medical insurance claims, immigration applications, and credit approvals are increasingly supported by machine learning systems. In general, these systems are not fully automated decision makers. Instead, automated models are embedded inside structured human review pipelines, where cases can be autonomously resolved or escalated to one or more human reviewers depending on factors including uncertainty, risk, and institutional constraints [1], [2].

This problem is called *selective prediction*, where a model is allowed to abstain on inputs where it has low confidence and defer to an expert human decision-maker [3], [4]. Previous works have studied selective classification and abstention, describing tradeoffs between prediction risk and coverage. Works such as [4], [5] have developed confidence-based deferral policies with strong theoretical guarantees. These approaches have been shown to be effective in applications where the alternative to automated prediction is a single, homogenous human expert.

However, real-world decision pipelines rarely conform to single-stage abstraction. In practice, human review is hierarchical, where cases can be routed to reviewers with differing levels of expertise, authority, and cost. For example, in insurance claim resolution, an application may be resolved automatically, reviewed by a junior examiner, escalated to a senior specialist, or forwarded to an administrative judge. Modelling deferral as a binary decision process does not always model this system well as it obscures important tradeoffs between accuracy and human resource allocation.

Previous work on algorithmic triage and human-AI collaboration has studied how automated systems can allocate cases between humans and models [6], [7], [8]. While these approaches acknowledge heterogeneous human expertise, they focus on two-stage settings, rarely on heuristic or fixed routing rules, or do not explicitly connect to the selective prediction framework. As a result, existing methods provide limited guidance for designing multi-stage pipelines under explicit budget constraints.

In this work, we introduce Human-Aware Routing Policies (HARP), a framework for selective prediction in multi-stage

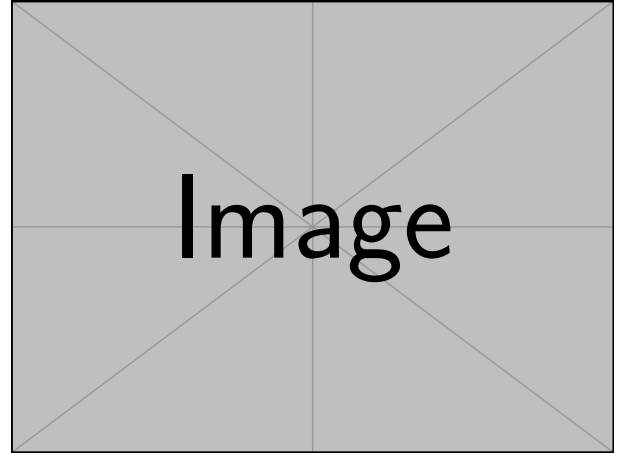


Fig. 1. Example figure caption.

human-AI decision pipelines. Rather than making deferral a binary action, HARP formulates decision making as a routing problem over a hierarchy of human reviewers with increasing costs, with the objective of minimizing system loss while also respecting human resource constraints.

We study this framework in the context of structured application review pipelines motivated by Social Security Disability Insurance (SSDI) claim resolution, where decisions are sequential, resource constrained, and subject to high institutional stakes. This setting shows the limitations of single-stage selective prediction and motivates the need for routing-aware decision policies.

Our work makes the following contributions:

- contrib1
- contrib 2
- contrib 3

II. PROBLEM FORMULATION

A. Single Stage Policies

We specifically study human-AI collaborative decision making in structured application review pipelines, where an automated system can either issue a decision or defer the decision to a human reviewer at some cost.

Let

$(x, y) \sim D$, where D is the real world distribution over applications and ground-truth decisions

$x \in \mathcal{X}$, where x is an application

$y \in \mathcal{Y}$, where y is the ground truth decision

We consider a predictive model $f : \mathcal{X} \rightarrow \mathcal{Y}$ and a deferral policy $r : \mathcal{X} \rightarrow \{0, 1\}$, where

$$\begin{aligned} r(x) = 0 & \quad \text{when the policy chooses to predict} \\ & \quad \text{autonomously} \\ r(x) = 1 & \quad \text{when the policy defers to a human} \\ & \quad \text{for a cost } c_h > 0. \end{aligned}$$

When $r(x) = 0$, the automated system incurs cost $\mathcal{L}_1$, which is given by

$$\mathcal{L}_1(f) = \ell(f(x), y),$$

where ℓ is by convention 0-1 loss or cross-entropy.

When $r(x) = 1$, the system defers to a human for a human cost $c_h > 0$, assuming that the human produces the correct decision. Therefore, the total system loss can be modeled as the expectation of a function of f and r :

$$\begin{aligned} \mathcal{L}(f, r) = \mathbb{E}_{(x, y) \sim D} & \left[\ell(f(x), y) \cdot \mathbf{1}\{r(x) = 0\} \right. \\ & \left. + c_h \cdot \mathbf{1}\{r(x) = 1\} \right] \end{aligned} \quad (1)$$

In this selective prediction model, when the policy $r(x) = 0$, f predicts autonomously. Over time, as $r(x)$ continues to be 1, c_h will increase. Therefore, we can define a coverage function of r as follows:

$$\text{cov}(r) = \mathbb{E}_{(x, y) \sim D} [\mathbf{1}\{r(x) = 0\}], \quad (2)$$

which describes the fraction for which the policy r chooses for f not to abstain.

When $r(x) = 0$, $f(x)$ is evaluated and we can define the conditional risk as the loss ℓ between $f(x)$ and y for a given $(x, y) \sim D$. More formally, this is:

$$\text{risk}(f, r) = \mathbb{E}_{(x, y) \sim D} [\ell(f(x), y) \mid r(x) = 0], \quad (3)$$

which shows the conditional risk when f autonomously makes the decision.

Based on this coverage function and the conditional risk $\mathcal{L}$, we can define a coverage-constrained optimization problem as follows:

$$\min_{f, r} \mathbb{E}[\ell(f(x), y) \mid r(x) = 0], \quad \text{s.t. } \text{cov}(r) \geq \kappa, \quad (4)$$

where κ is the constraint for the fraction where $r(x) = 0$.

Therefore, we can apply a confidence-based threshold defined on the confidence from the output of f , which is traditionally just the max of the output of the sigmoid or softmax function if f is a neural network. Thus, we let:

$$p_\theta(y|x) \quad \text{be the predictive distribution of} \\ f \quad (5)$$

$$\begin{aligned} \text{conf}(x) = \max_y p_\theta(y|x) & \quad \text{be defined as the confidence} \\ & \quad \text{of a given application-decision} \\ & \quad \text{pair.} \end{aligned} \quad (6)$$

We can then threshold the policy r on the confidence function for a hyperparameter τ as follows:

$$r_\tau(x) = \begin{cases} 0 & \text{if } \text{conf}(x) \geq \tau, \\ 1 & \text{if } \text{conf}(x) < \tau \end{cases} \quad (7)$$

B. Extension to Multi-Stage Policies

The optimization problem in Eq. (4) is applicable when we have a binary decision of using policy f or abstaining and deferring to an arbitrary human reviewer. However, when we have a hierarchical set of k human reviewers $H = \{0, 1, \dots, k\}$, each with increasing costs c_{h_k} , we have c_{h_k} increasing when $k \geq 1$.

For the purposes of this paper, single-stage refers to $|H| = 1$, and multi-stage is when $|H| > 1$. We extend the single stage-policy to multi-stage. In this case, we make a trivial change to the minimization problem in Eq. (4) and the initial routing policy r . Let

$$\begin{aligned} r_m : \mathcal{X} \rightarrow \{0, 1, \dots, k\} & \quad \text{be the multistage routing policy,} \\ & \quad (8) \end{aligned}$$

$$\begin{aligned} \ell_s(x) & \quad \text{be the loss of sending } x \text{ to} \\ & \quad \text{stage } s, \end{aligned} \quad (9)$$

$$\begin{aligned} B & \quad \text{be the total budget constraint} \\ & \quad \text{for the human resources avail-} \\ & \quad \text{able.} \end{aligned} \quad (10)$$

We assume $\ell_0(x) = \ell(f(x), y)$ corresponds to the automated model loss, and $\ell_s(x) = c_{h_s}$ for $s \geq 1$, assuming human reviewers produce correct decisions.

Therefore, the constrained optimization problem we must solve is as follows:

$$\min_{r_m} \mathbb{E}[\ell_{r_m(x)}(x)], \quad \text{s.t. } \mathbb{E}[c_{h_{r_m(x)}}] \leq B. \quad (11)$$

C. Stochastic Routing Policies

In practice, stochastic routing allows for calculating gradients in optimization, where the system learns a probabilistic policy over decision stages. Rather than deterministically assigning an application x to a single stage, we define a routing distribution $\pi_\phi(s \mid x)$ over stages $s \in \{0, 1, \dots, k\}$, parameterized by ϕ . For each input, a stage is sampled according to this distribution, allowing the routing policy to express uncertainty and to smooth decisions near stage boundaries.

Under this formulation, the expected system loss is given by

$$\mathbb{E}_{(x, y) \sim D} \left[\sum_{s=0}^k \pi_\phi(s \mid x) \ell_s(x) \right], \quad (12)$$

with an associated expected human cost

$$\mathbb{E}_{x \sim D} \left[\sum_{s=0}^k \pi_\phi(s \mid x) c_{h_s} \right]. \quad (13)$$

Therefore, at training time the system is optimized stochastically, but during inference it is deterministic.

III. BACKGROUND AND RELATED WORK

IV. MATERIALS AND METHODS

Describe the data, materials, experimental setup, or methodological framework. This section should be detailed enough to allow reproducibility.

A. Data or Materials

Describe sources, collection procedures, and properties.

B. Methodology

Explain analytical, computational, or experimental methods used.

V. RESULTS

Present results objectively, using tables and figures where appropriate.

VI. DISCUSSION

Interpret the results, explain trends, limitations, and unexpected outcomes. Relate findings back to the motivation and prior work.

VII. CONCLUSION AND FUTURE WORK

Summarize key contributions and outline potential directions for future research.

AUTHOR NOTE

The authors declare no competing interests.

AUTHOR CONTRIBUTIONS

Describe each author's contribution to the work.

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